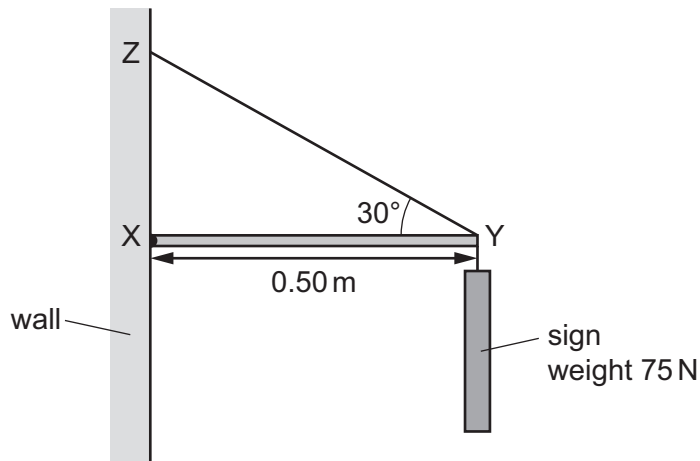


- 11** A shop sign weighing 75 N hangs from a frame attached to a vertical wall.



The frame consists of a horizontal rod XY and a rod YZ that is at an angle of 30° to the horizontal. Rod XY is attached to the wall by a hinge at X and has length 0.50 m. Assume that the weights of the rods are negligible.

What is the horizontal force exerted by the wall on rod XY?

- A** 0 N **B** 43 N **C** 130 N **D** 150 N

- 12** A student takes measurements to calculate the density of a liquid in a beaker